

MLB Player Performance Analytics

Where Traditional Stats and Statcast Metrics Agree, Disagree,
and What It Means for Player Evaluation

AJ Schoolcraft | ISBA 4715 Capstone | Spring 2026

Target Role: Baseball Operations Analyst, San Francisco Giants

Traditional Stats Systematically Undervalue Elite Contact Quality

Descriptive: What happened across the league?

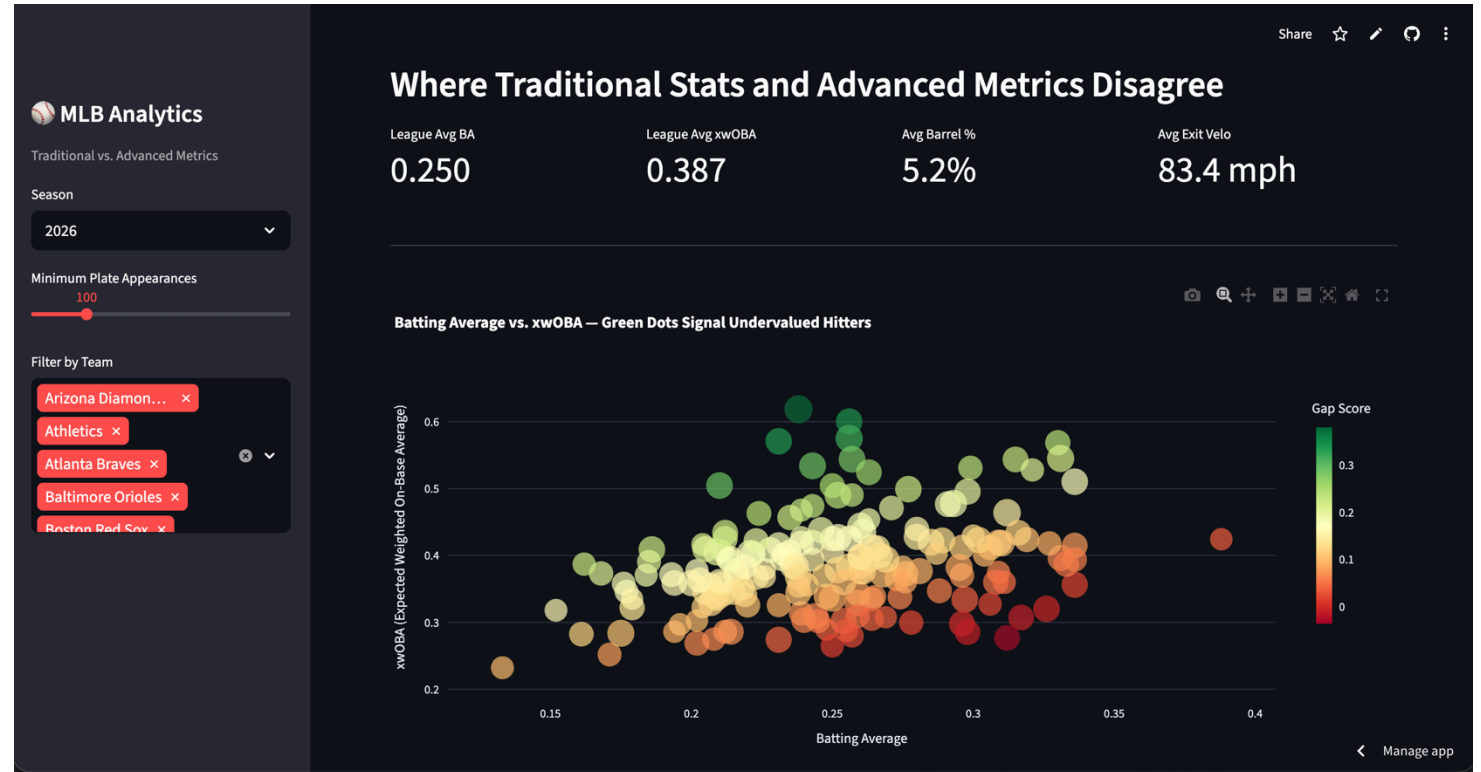
Players with barrel rates above 10% outperform their batting average by ~45 points in xwOBA

Barrels produce a .684 BA and 2.236 SLG vs. .292/.368

for non-barreled contact (2026 data)

The scatter plot reveals a clear cluster of 'undervalued' hitters — high xwOBA, low batting average

These gaps are not random — they correlate with measurable batted-ball quality (exit velocity, barrel %)



High Exit Velocity + Low BABIP = Unlucky, Not Untalented

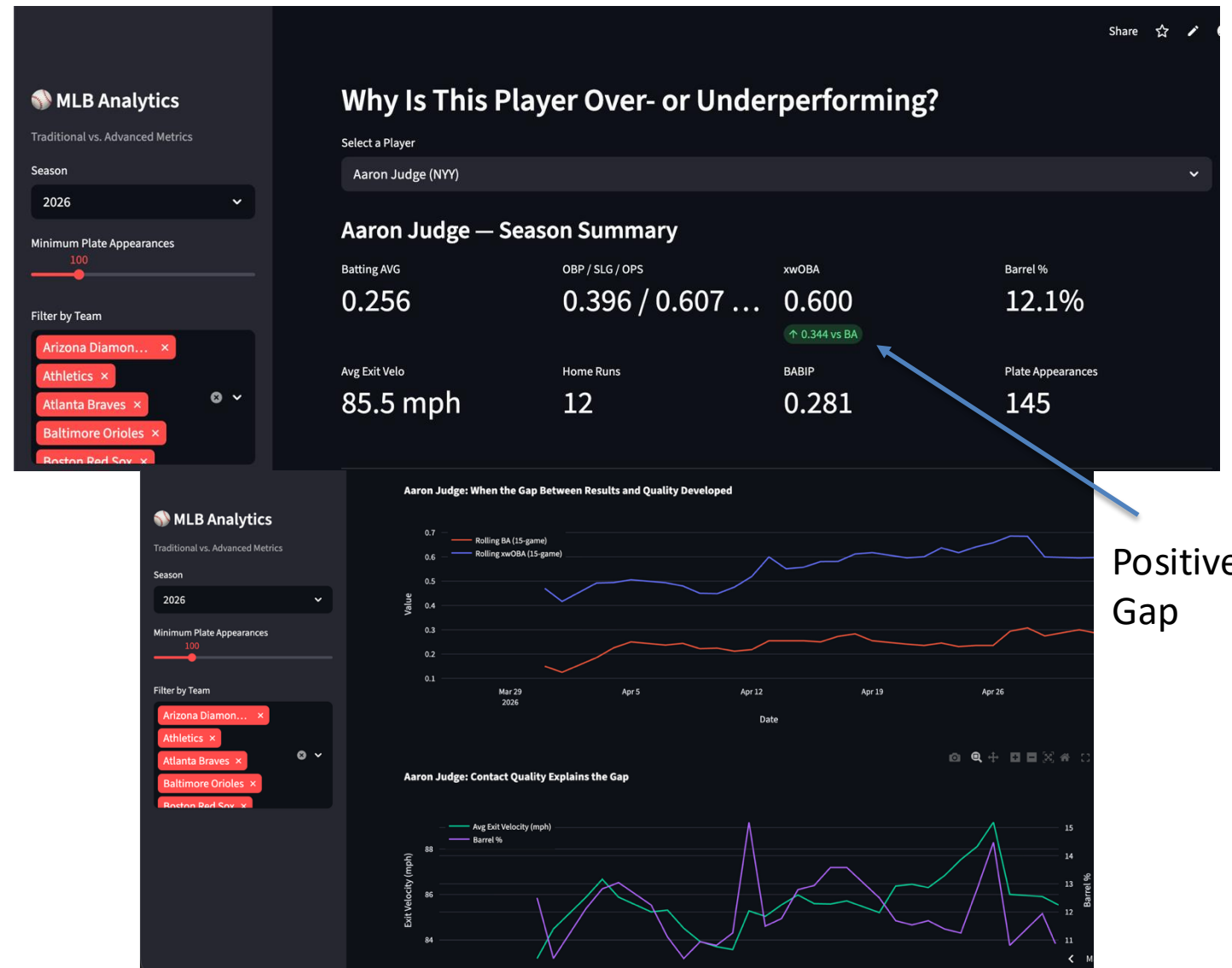
Diagnostic: Why do results diverge from underlying quality?

The gap between xwOBA and batting average is largest for players with high exit velocity but below-average BABIP

Rolling 15-game trends reveal when the gap develops — often after a streak of hard-hit balls finding fielders

Contact quality charts (barrel % + exit velo) confirm the hitter's process is sound even when results lag

BABIP regresses toward ~.300 over time — players with low BABIP and high barrel rates are buy-low candidates



Recommendation: Target Undervalued Hitters for Acquisition

Action → Expected Outcome

ACTION: Target players with top-quartile barrel rates ($>10\%$) and below-average BABIP ($<.280$)

OUTCOME: Expected regression to the mean projects 15–25 point batting average increases

VALUE: These players are available below market value because traditional stats understate their talent

Decision Framework

1. Screen: Barrel % $> 10\%$, BABIP $< .280$, PA > 100
2. Diagnose: Check xwOBA vs. BA gap score (positive = undervalued)
3. Confirm: Review rolling trend — is contact quality sustained or fluky?
4. Act: Acquire before BABIP regression corrects the market price

End-to-End Pipeline: From Raw APIs to Actionable Dashboard

Architecture & Tech Stack

Extract: MLB Stats API + Pybaseball (Statcast)
pull player stats, game logs, and batted-ball data

Load: Python scripts write to Snowflake RAW schema
via snowflake-connector-python

Transform: dbt builds staging views and mart tables
(star schema: 2 facts + 4 dimensions)

Present: Streamlit dashboard with Plotly charts
deployed to Community Cloud

Knowledge Base: 16 sources scraped from FanGraphs,
Baseball Savant, BP → 4 synthesized wiki pages

Orchestrate: GitHub Actions runs daily ELT
and weekly knowledge base re-scrape

